



Lesson 5 – Design for Good

Sustainability, societal impact and equity

Teacher-only slide – Lesson Overview

Lesson Preparation



- Print off Design for Good Worksheet, one per group of 3-4 is appropriate (total 4 pages)

Design for Good



Perception

We could perceive Blossom to be lots of different things, and how we want Blossom to be perceived influences our design.

First, let's plan how we want to prioritise our different design considerations.

We want Blossom to look...	Ranking (1-4)
Smart	
Kind	
Helpful	
Aesthetic	

Research some materials, give three suggestions of materials you perceive as whatever your most important design consideration is

Material	Score X/10
Smart	
Kind	
Helpful	

Design for Good



Sustainability

We want our skin to be sustainable, but sustainability can be broken down into a few categories.

Let's do some research on each of our materials and begin to score them for sustainability.

Material	Where did it come from?	Is it recyclable?	Is it durable (how long will it last)?	Score (X/10)

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Important instruction



- Students will choose a material to design Blossom's skin in this lesson. The idea is that they will source that material from home to design next week.
- At the end of the lesson get students to plan in their groups who will be sourcing what, if multiple materials. The lesson is designed such that materials are sustainable and equitable, they should be free and repurposed. If any groups run into trouble sourcing materials they are welcome to use the recycled shopping bags we provided.

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Lesson 5 - Overview



1. **Lesson 4 Recap** - 5 min
 2. **Design factors** – 15 min
 - 2.1 Perception
 - 2.2 Sustainability
 - 2.3 Equity
 3. **Desing for Good worksheet** – 30 min
 4. **Interacting with Blossom** - 30 min
 5. **Survey** - 5 min
 6. **Summary** - 5 min
- Total: 90 min

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Lesson 5 – Learning objectives



After this lesson, students will be able to:

- Explain the environmental benefit of using recycled materials in design
- Describe what inclusive design means and why it matters
- Identify how a robot's appearance influences human perception and trust
- Reflect on how design choices affect who can access and benefit from technology
- Evaluate whether a design decision is equitable by considering who is included or excluded
- Recognise that good design involves values and social responsibility, not just function

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Lesson 5 - Student success criteria



- I can explain why using recycled materials is a more sustainable design choice
- I can describe what equity means in the context of design and technology
- I can give an example of a design that includes some people and excludes others
- I can explain how Blossom's appearance might change the way someone feels about it
- I can reflect on the design choices I made and explain the values behind them
- I can describe what "design for good" means in my own words

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Lesson 4 Recap



- **Instruction note:** Get students to come up with some examples of AI again, maybe discussed if they noticed any examples of AI in day to day life since last week.
- **Sample script:** *“AI is when computers are programmed to do things that would normally require thinking (like humans do). It learns from patterns in data, just like how we learn from our experiences. AI can make mistakes too, it’s important to check the answers that AI gives us.”*

Let's remember what we learned in Lesson 4!



- What is AI?
 - Give 3 examples of AI we come across everyday
- Why do we need to check the answers AI gives us?
- AI predicts based on patterns, it doesn't think or understand things like we do!

What is perception?



- **Sample script:** *“Perception is how we view things, or how we feel about things when we see them. Our perception of Blossom can change based on how we design the skin and what Blossom looks like. On the next slide we are going to see four different skins Outback Robots team dressed Blossom in.”*

What is perception?



- **Perception = how we view things**
- 🧐 How something LOOKS changes how we feel about it
- 🤖 Blossom's skin changes how we perceive it
- ***Same robot — different skin — different reaction!***

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Skins



- **Instruction note:** Play an interactive voting game, for each question get the students to move to a corner of the room for which skin they perceive as the smartest, best and which they would treat the most carefully. Maybe get a student to discuss why each time.

Skins



Skin A

[Add photo here]

Skin B

[Add photo here]

Skin C

[Add photo here]

Skin D

[Add photo here]

 **Think about...**

 Which Blossom seems the smartest?  Which looks the best?

 Which would you treat most carefully?

Perception changes



- **Instruction note:** This can be more of an ‘informal chat’. Discuss with the students who changed their view and why. Was anyone’s vote influenced by peers. Notice our perception of which Blossom was ‘best’ depends on what the question is, smart, best looking or the one we’d treat most carefully.

Perception changes



- **Our perceptions change!**
-  Not everyone agrees — and that's ok!
-  Peers can even change our minds
- ***Same Blossom — same function — materials matter!***

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Sustainability



- **Sample script:** *"Sustainability is about how we take care of our world, making sure the choices we make today don't cause problems for the future. Sustainable materials are usually spoken about as materials we can recycle, are durable and easy to get."*
- *When we design Blossom's skin, we need to think about where the materials come from, how they're made, and what happens to them when we're done."*

Sustainability



- **Sustainability = caring for the world's future**



Recyclable

can it be used again?



Durable

will it last?



Easy to get

available everywhere?

Sustainability



- Consider the 4??? Blossom skins
 - What is each made out of?
 - Is this new or reused material?
 - Are the materials recyclable?
 - How durable are they?
 - Could the material be used again?

Teacher-only slide – Instruction Guide

Equity



- **Sample script:** *“Equity is providing everyone with equal opportunities to access something. Think about things that are low cost, available in all locations (abundant) and easy to work with.*
- *If we asked every student in this class to bring in their own material from home, would that be fair? Would everyone be able to? Remember to be equitable it has to cover all of the points, cost, availability and easy to use”*

Equity



- **Equity = equal access for everyone**
- Every Australian school must be able to use the material!



Cost

gold sheets vs plastic
bags



Availability

old t-shirts vs specialty
fabric



Ease of use

cut & staple vs knitting

Why materials matter



- **Sample script:** *“For this activity we are going to pretend we are engineers for Outback Robots. We are going to design a skin for Blossom that can be used by every school in Australia. We need to think about the three big questions on the slide.”*
- *First , perception. How will people feel when they see Blossom for the first time? Will it look friendly? Smart? Professional? The way something looks affects how people react to it.*
- *Second, sustainability. If every school in Australia builds this skin, that's a LOT of material. We need to ask: is it kind to the environment? Can it be recycled or reused?*
- *Equity. Can a school in remote areas get the same materials as a school in Adelaide? Our design has to work for everyone, not just schools with big budgets and fancy equipment.”*

Your turn



- **You are the head engineer for Outback Robots!**

- Design a skin for every school in Australia

- Consider:



Perception



Sustainability



Equity

Design for Good Worksheet



- **Instruction note: Get students to work in small groups choosing a material for Blossom's skin. Encourage them to do their own research on three different materials they might see as appropriate to design a skin for Blossom. There are no right or wrong answers.**
- **Get each group to do a brief presentation/discuss their findings and suggest which material they would use to design Blossom's skin, remembering it's thinking about the use being every Australian school!**

Your turn



- Split into groups and design a solution
- Remember your solution needs to be
 - Well perceived (for your intended use)
 - Sustainable
 - Equitable
- Let's use the Design for Good worksheets to help us plan

Questionnaire



Summary



- Give a brief presentation on what material you chose and why!